Welcome to Linux Distributions. After watching this video, you will be able to: Describe what a Linux distribution is. Differentiate between some common Linux distributions. And identify the use cases of some popular Linux distributions. So, what is a Linux distribution? A Linux distribution is a specific flavor of the Linux operating system. It may also be referred to as a distro. All Linux distributions must use the Linux kernel. This is the core component of the Linux operating system that enables the system to properly use the computer's hardware. And today, there are hundreds of Linux distros, each tailored for a specific audience or task. Okay, so what differentiates Linux distros from each other? Each Linux distro includes a unique set of default utilities that are part of the operating system, such as commands and applications that come prepackaged with the distribution. Each distro has its own graphical user interface, or GUI, that you use to interact with the operating system. Each distro supports a specific set of commands that you can use in a shell, a window for entering and receiving output from commands. And lastly, each distro provides differing levels of support. It could be developed and managed as a community-backed project or maintained by a commercial enterprise. And it could be a long-term support version, or LTS, or a rolling release, where stable package updates are released on a rolling schedule. Debian is one of the earliest-rooted distros, with the first release in 1993 (named version 0.01) and the first official, stable release in 1996 (version 1.1). It is known for being stable, reliable, and fully open source. It supports many computer architectures (or types of hardware). These features result in Debian being highly regarded in the server space. And furthermore, Debian is the largest community-run distro currently available. Another popular distro is Ubuntu. This is also an early-rooted distro, with its first official release in 2004. Ubuntu is Debian-based, meaning that it’s built on top of Debian and uses a lot of the same tools as the Debian OS. Ubuntu is developed and managed by Canonical. And Ubuntu has three official editions: Desktop (for personal computers, laptops, and workstations) Server (for simple file servers or multinode clouds) and Core (for the Internet of Things). Red Hat Linux, like Debian, is a "core" Linux distro, meaning that it is not derived from another Linux distro. Red Hat is known for being stable, reliable, and fully open source. It’s managed by Red Hat, an IBM subsidiary. Today it ships as Red Hat Enterprise Linux (or RHEL), an edition focused entirely on enterprise customers. Next is Fedora, which is known as a stable operating system that supports many architectures. It’s also very reliable and secure, offering unique firewall and security features. It’s actively developed, with a large and growing community. And it’s sponsored by Red Hat. Consequently, Red Hat uses much of Fedora’s code base (after testing, of course). Another Linux distro is SUSE Linux Enterprise. Also known as SLE, SUSE Linux Enterprise is available in two editions: Server, or SLES, and Desktop, or SLED. It supports many architectures, such as ARM for Raspberry Pi, and uses the SUSE Package Hub, which enables users to install packages that aren’t officially part of SLE. And it's maintained by the small German open-source software company, SUSE. And finally, we have Arch Linux. The unique, do-it-yourself approach of Arch Linux allows users to customize every part of their system. This makes it highly configurable, a freedom that makes Arch very popular. But it also means that you need a strong understanding of Linux and system tools in order to use Arch effectively. And because Arch isn’t focused on stability in the same way as other distros, it has easy access to the newest software, which has little guarantee of being completely stable. In this video, you learned that: Linux distros can be differentiated by their user interfaces, their shell applications, and how the operating system is supported and built. The design of a Linux

distro is catered toward a specific audience. Debian is highly regarded in the server space for its stability, reliability, and for being open source. Red Hat Enterprise Linux, an IBM subsidiary, is focused completely on enterprise customers. And SUSE Linux Enterprise supports many architectures, such as ARM for Raspberry Pi.